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Control of stability and structure of nucleic acids using cosolutes

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ABSTRACT

The stabilities, structures, and functions of nucleic acids are responsive to surrounding conditions. Living cells contain biomolecules, including nucleic acids, proteins, polysaccharides, and other soluble and insoluble low-molecular weight components, that occupy a significant fraction of the cellular volume (up to 40%), resulting in a highly crowded intracellular environment. We have proven that conditions that mimic features of this intra-cellular environment alter the physical properties affect the stability, structure, and function of nucleic acids. The ability to control structure of nucleic acids by mimicking intra-cellular conditions will be useful in nanotechnology applications of nucleic acids. This paper describes methods that can be used to analyze quantitatively the intra-cellular environment effects caused by cosolutes on nucleic acid structures and to regulate properties of nucleic acids using cosolutes.
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1. Introduction

Controlled formation of structures by nucleic acids has enormous potential in the fields of nanobiotechnology and biomedical technology. Nucleic acids can undergo structural transitions from the canonical duplex structure to non-canonical structures such as a triplex, a quadruplex, i-motif, or cruciform in response to environmental stimuli. Such stimuli include pH, [1–3] small molecules (e.g., cosolutes and ions), [4–7] macromolecules (e.g., oligonucleotides and proteins), [8] and electrical signals. [9] Researchers have taken advantage of the switching properties of nucleic acid structures to develop nucleic acid-based sensors, logic devices, circuits, and drugs. We and other researchers also have developed logic devices that respond to input molecules (e.g., pH and cations) that monitor the change in DNA structures from G-quadruplexes and i-motifs to duplexes, and these are of considerable interest in DNA computing. Thus, control of the structures formed by nucleic acids is of interest in diverse research fields including medical, pharmaceutical, and material sciences.

The following five factors play major roles in determining the stabilities and structures of nucleic acids: hydrogen bonding, base stacking, conformational entropy, water binding, and cation binding. Hydrogen bonding, base stacking, and conformational entropy are determined by sequence of nucleic acids [10–13]. In contrast, extents of water and cation binding are determined by conditions surrounding the nucleic acid such as cosolute and cosolvent conditions [14–19]. Thus, a convenient method for control of the structures formed by nucleic acids and the stabilities of those structures is to change the surrounding conditions.

In living cells, organelles such as nuclei, the endoplasmic reticulum, and the mitochondria are present (Fig. 1a). [20,21] Moreover, biomolecules such as nucleic acids, proteins, polysaccharides, and other soluble and insoluble low-molecular weight components occupy a significant fraction of the cellular volume (up to 40%) (Fig. 1b) [21–23]. The effect of high concentrations of biomolecules is referred to as molecular crowding. Under the crowded conditions, the chemical potentials and activity coefficients of biomolecules are changed due to the excluded volume effect. Moreover, the physical properties of intra-cellular conditions such as osmotic pressure, viscosity, and the dielectric constant are different from those of dilute solutions [23]. We have investigated the effect of intra-cellular conditions induced by cosolutes on the structures and stabilities of nucleic acids: Cosolutes, especially cosolutes with low-molecular weight (below about 200), change the physical properties of solution and perturb the water and cation binding to nucleic acids [23–27]. This paper describes the methods for quantitative analysis for water and cation binding to DNA in the presence of cosolutes and describes how one can take advantage of the properties of cosolutes to regulate stabilities and structures of nucleic acids.

2. Sample preparation

2.1. Preparation of nucleic acids

Oligodeoxyribonucleotides are suitable for quantitative studies because association and dissociation of the structures usually

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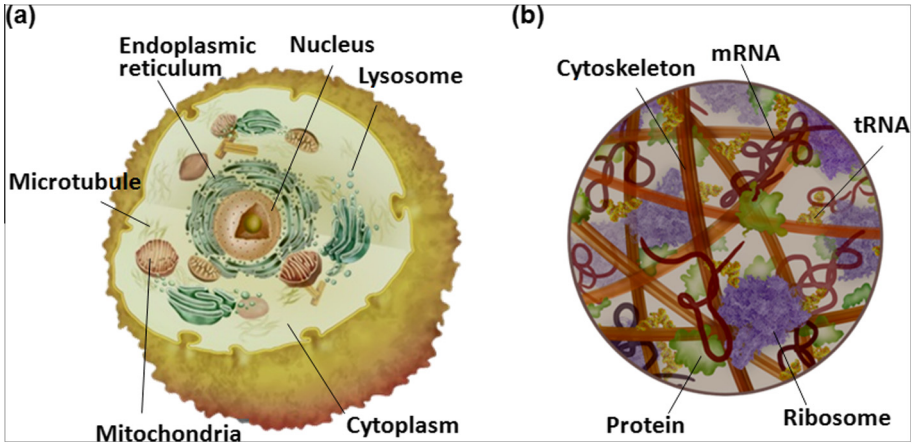

Fig. 1. Schematic structures of a highly crowded environment in cell resulting from the presence of (a) organelles and (b) macromolecules.

occur through two-state transitions; this property makes it possible to evaluate the thermodynamic parameters of the conformational transition [24,25,28]. In addition, many short oligonucleotides adopt the same structures in both dilute and in molecular crowding conditions [24,25,27,29]. In contrast, long nucleic acid strands tend to fold into compact and aggregated structures under crowding conditions that differ from those adopted in dilute solution [24,30]. Oligonucleotides are synthesized using a solid-phase phosphoroamidate procedure on an automated nucleic acid synthesizer. The synthesized oligonucleotides are purified by reversed-phase HPLC (typically using a C18 column with a methanol gradient). The purity of the purified oligonucleotides may be analyzed by HPLC or mass spectroscopy. Oligonucleotides of less than 95% purity should not be used for quantitative data analysis. HPCL-purified oligonucleotides can be purchased from a number of companies (e.g., Wako Pure Chemical, Tokyo Chemical Industry, and Sigma–Aldrich Japan). Single-strand concentrations of DNA and RNA oligonucleotides are determined from the absorbance measured at 260 nm at 90 °C using the single-strand extinction coefficients calculated from the mononucleotide and dinucleotide data based on the nearest-neighbor approximation model [31].

2.2. Preparation of crowding solution

Various compounds have been used as cosolute molecules to mimic the conditions that occurs naturally in cells. The cosolute concentrations are often represented as weight percent per volume and the cosolutes are generally used at 5–40 wt%. Cosolutes used to mimic cellular crowding should meet the following criteria: *(1)* Molecules should be basically inert (i.e., the cosolutes should not interact directly with nucleic acids), *(2)* cosolutes should be soluble in water (to at least several weight percent), and *(3)* in the case of large cosolutes, a range of polymer sizes should be available [20,22,23,32].

The commonly used large cosolutes are poly(ethylene glycol) (PEG), dextran, and Ficoll. PEG is often used because different molecular weight PEGs are available [23]. Proteins such as albumin, hemoglobin, and lysozyme are also utilized as crowding cosolutes [32]. Commonly used small cosolutes are ethylene glycol (EG), PEG 200 (PEG with average molecular weight of 200), alcohols, glycols, amino acids, and betaine [17,24,33]. These cosolutes have been used in experimental studies to study the equilibria and kinetics of many biochemical reactions. These cosolutes can be purchased from many commercial sources (e.g., Wako Pure Chemical, Tokyo Chemical Industry, and Sigma–Aldrich Japan).

2.3. Thermodynamic analysis for formation of nucleic acid structures

Methods to obtain the thermodynamic parameters of enthalpy (ΔH°), entropy (ΔS°), and free-energy changes at 25 °C (ΔG°_{25}) for the formation of a nucleic acid structures are described below; DNA duplex formation is taken as an example. Data are typically analyzed with a two-state model, which assumes that each strand is either completely paired or unpaired. The equilibrium for the duplex formation is represented as either a self-complementary or non-self-complementary association as follows [34,35]:

$$2A \rightleftarrows A_2 \tag{1}$$

$$B + C \rightleftarrows B \cdot C \tag{2}$$

where A, B, and C indicate the single strands of DNA and A_2 and $B \cdot C$ indicate the double-stranded DNA.

The equilibrium for a unimolecular transition, for example, of a hairpin, is represented as:

$$D_{ss} \rightleftarrows D_f \tag{3}$$

where D_{ss} and D_f indicate the single-stranded (unfolded) and folded DNA structures, respectively.

For self-complementary (Eq. (1)) or non-self-complementary equilibria (Eq. (2)) with equal concentrations of B and C, the observed equilibrium constant is given by:

$$K_{obs} = (\alpha/2)/(C_t/s)(1-\alpha)^2 \tag{4}$$

C_t is the total strand concentration:

$$C_t = [A] + 2[A]\ (\text{self} - \text{complementary duplex}) \tag{5}$$

$$C_t = [B] + [C] + [B \cdot C]\ (\text{non-self} - \text{complementary duplex}) \tag{6}$$

s has a value of 1 for self-complementary duplexes and 4 for non-self-complementary duplexes. α is the fraction of strands in a duplex. For a unimolecular transition,

$$K_{obs} = \alpha/(1-\alpha) = [D_f]/[D_{ss}] \tag{7}$$

where

$$\alpha = [D_f]/([D_{ss}] + [D_f]) \tag{8}$$

Fig. 2 shows a UV melting curve for the self-complementary duplex (5′-ATGCGCAT-3′). At low temperatures, the strands are in duplex form and the absorbance is low. As the temperature is increased, the duplex dissociates into single strands. The difference in absorbance between duplex and single strands is referred to as hyperchromicity. For self-complementary or

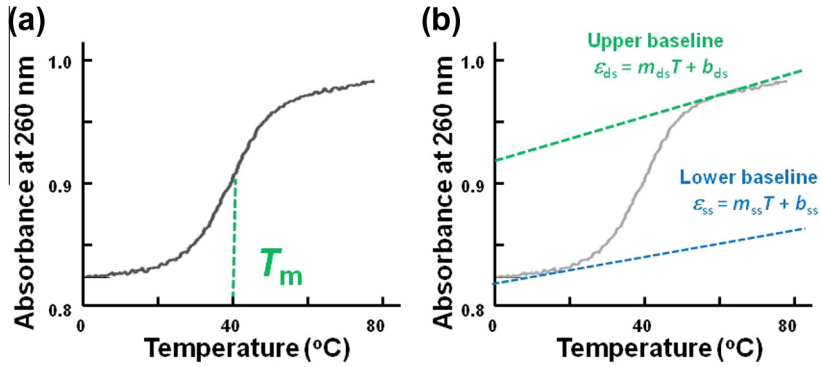

Fig. 2. UV melting curves of the self-complementary duplex (5′-ATGCGCAT-3′) at 5 μM strand concentration with (a) T_m value and (b) upper and lower base lines of the UV melting curve. Upper and lower baselines can be represented as $\varepsilon_{ds} = m_{ds}T + b_{ds}$ and $\varepsilon_{ss}=m_{ss}T + b_{ss}$, respectively, where ε_{ds} and ε_{ss} indicate the absorbance for the single-stranded and double-stranded DNA, respectively. The m_{ds} and b_{ds} or m_{ss} and b_{ss} represent the slope and intercept of the upper base line or lower base line for the UV melting curve, respectively.

non-self-complementary duplexes with equal concentrations of each strand, the melting temperature, T_m (in degrees kelvin), is the point at which the concentrations of strands in duplex and in single strands are equal (Fig. 2a). The steepness of the transition indicates the cooperativity of the transition. The width and maximum of first derivative of the melting curve can also indicate the cooperativity and melting temperature, although the peak of the derivative curve only occurs at the T_m only when the transition is unimolecular [36]. T_m is most accurately measured by fitting the lower and upper baselines. The melting temperature is measured at several concentrations over a 100-fold range and then plotted versus the concentration in a van't Hoff plot. The van't Hoff equation relates the T_m (in degrees kelvin), C_t, $\Delta H°$, and $\Delta S°$:

$$T_m^{-1} = R\ln(C_t/s)/\Delta H° + \Delta S°/\Delta H° \quad (9)$$

where R is the ideal gas constant, 1.987 cal K^{-1} mol^{-1} or 8.3145 J K^{-1} mol^{-1}. The slope of the van't Hoff plot gives the $\Delta H°$, and the y-intercept gives the ratio of $\Delta H°$ to $\Delta S°$. The free energy and equilibrium constant at any temperature can then be calculated using Gibb's relation:

$$\Delta G°_T = \Delta H° - T \cdot \Delta S°,\ K_{obs} = e^{-\Delta G°/RT} \quad (10)$$

To increase the accuracy of these parameters, data analysis can be performed by curve fitting as shown below. When the ratio of the double-stranded DNA is represented by α, absorbance (A) at a temperature (T) is calculated as:

$$\varepsilon(T) = A(T)/(l \cdot C_t) = \varepsilon_{ds} + 2(1-\alpha)\varepsilon_{ss} \quad (11)$$

where ε_{ds} and ε_{ss} indicate the absorbance for the single-stranded and double-stranded DNA, respectively; l and C_t represent the length of the light pass (or the path length of the cuvette used) and the total concentration of DNA strands, respectively. The ε_{ds}, ε_{ss} and observed equilibrium constant (K_{obs}) for the duplex formation can be represented as follows with the assumption that absorbance is directly proportional to temperature:

$$\varepsilon_{ds} = m_{ds}T + b_{ds} \quad (12)$$

$$\varepsilon_{ss} = m_{ss}T + b_{ss} \quad (13)$$

$$K_{obs} = \exp(-\Delta H°/RT + \Delta S°/R) \quad (14)$$

where R is the gas constant and m_{ds} and b_{ds} or m_{ss} and b_{ss} represent the slope and intercept of the upper base line or lower base line for the melting curve of a duplex dissociation (Fig. 2b), respectively. The six variables (ε_{ds}, ε_{ss}, b_{ds}, b_{ss}, $\Delta H°$, and $\Delta S°$) can be calculated by the curve fitting using Eqs. (11)–(14). These calculations, which are performed using a PC equipped with the curve fitting software such as IGOR Pro, KaleidaGraph, or ORIGIN, derive the thermodynamic parameters from the shape of the melting curve.

Estimated errors in the thermodynamic values ($\sigma_{\Delta H°}$, $\sigma_{\Delta S°}$, and $\sigma_{\Delta G°25}$) derived from the curve fitting procedure are calculated from the standard deviations among data points of each melting curve at different C_t values. Estimated errors in $\Delta H°(\sigma_{\Delta H°})$ and $\Delta S°(\sigma_{\Delta S°})$ obtained from the T_m^{-1} versus $\ln(C_t/s)$ plots are calculated from the linearity of the plots, and those for $\Delta G°T$ ($\sigma_{\Delta G°25}$) are calculated using Eq. (15):

$$(\sigma_{\Delta G°25})^2 = (\sigma_{\Delta H°})^2 + T^2(\sigma_{\Delta S°})^2 - 2 \cdot T(R_{\Delta H°,\Delta S°})\sigma_{\Delta H°}\sigma_{\Delta S°} \quad (15)$$

where $R_{\Delta H°,\Delta S°}$ is the correlation coefficient between $\Delta H°$ and $\Delta S°$. The final thermodynamic parameters are the average values obtained from curve fitting and the T_m^{-1} versus $\ln(C_t/s)$ plots. The errors of $\sigma_{\Delta H°}$, $\sigma_{\Delta S°}$, and $\sigma_{\Delta G°25}$ are usually within several percent of each thermodynamic value.

3. Theory for controlling the structures and stabilities of nucleic acids by molecular crowding

3.1. Determination of water activity

Water activity is determined by measurement of vapor phase or freezing point depression using an osmometer under the assumption that cosolutes do not directly interact with nucleic acids [21,24]. Although the osmotic pressure values for most cosolutes can be measured using a vapor phase osmometer, the osmotic pressure of cosolutes (cosolvents) with boiling points less than 100 °C should be obtained using a freezing point depression osmometer [33]. The freezing point depression osmometer is unsuitable for osmotic pressure measurements of high-molecular weight cosolutes because solutions containing these cosolutes sometimes do not freeze uniformly.

For example, the osmotic pressure for EG, PEGs of various molecular weights, and glycerol were obtained using a model 5520XR pressure osmometer (Wescor, USA) by the osmotic stressing method via the vapor phase. The values for 2-methoxyethanol, 1,2-dimethoxyethane, and 1,3-propandiol were obtained by freezing point depression osmometry using a Halbmikro osmometer Typ Dig. L (KNAUER, Berlin, Germany).

In calculating the water activity from the osmotic pressure, it is assumed that the temperature dependence of the water activity is small between 25 and 80 °C; this assumption has been confirmed experimentally [16,37]. The logarithm of water activities (a_w) is

calculated from the measured osmolality (mmol/kg) as osmotic pressure per unitary molarity: [16,19,38]

$$\Psi = (RT/M_w)\ln a_w \tag{16}$$

where ψ is the water potential, M_w is the molecular weight of water (0.018 kg/mol), R is the gas constant (8.314 J mol^{-1} K^{-1}), and T is the temperature in kelvin. Water potential is defined as the potential energy of water per unit volume relative to pure water in reference conditions and is obtained by dividing chemical potential of water by partial molar volume of water. The relationship between water potential and osmolality, assuming independence of the water potential on temperature, is calculated as follows: [39,40]

$$\Psi\ (in\ \mathrm{MPa}) = \mathrm{osmolality}\ (\mathrm{mmol\ kg^{-1}})/-400 \tag{17}$$

3.2. Quantitative analyses of nucleic acid stability under molecular crowding conditions

The molecular crowding effect induced by cosolute on nucleic acids is explained using the intra-molecular DNA structure formation. Formation of a DNA structure in a solution containing a cosolute (e.g., PEG) and a cation (e.g., Na^+) can be represented as follows: [24,25,41]

$$A_{ss} \rightleftarrows A_f + \Delta n_w H_2O + \Delta n_{cs} CS + \Delta n_{Na^+} Na^+ \tag{18}$$

where A_{ss} and A_f represent concentrations of single-stranded (unfolded) and folded structures, respectively, CS represents cosolute concentration, Na^+ represents the sodium ion concentration, and Δn_w, Δn_{cs}, and Δn_{Na+} represent the numbers of water molecules, cosolute molecules, and sodium ions released upon DNA structure formation, respectively. According to linkage concepts, [40] changes in the equilibrium constants for the above reactions are dependent on water activity and solute or ionic concentrations as determined by the following relationship:

$$-\mathrm{d}(\ln K_{obs}) = \left(\frac{\partial(\ln K_{obs})}{\partial(\ln a_w)}\right)\mathrm{d}(\ln a_w) + \left(\frac{\partial(\ln K_{obs})}{\partial(\ln a_{cs})}\right)\mathrm{d}(\ln a_{cs}) + \left(\frac{\partial(\ln K_{obs})}{\partial(\ln a_{Na^+})}\right)\mathrm{d}(\ln a_{Na^+}) \tag{19}$$

where K_{obs} is the observed thermodynamic equilibrium constant and a_w, a_{cs}, and a_{Na+} are the activities of water, the cosolute, and the sodium ion, respectively. It was shown by Wyman [42] that

$$\frac{\partial \ln K_{obs}}{\partial \ln a_i} = \Delta n_i\ (a_i \text{ is the activity of either water, cosolute, or } Na^+) \tag{20}$$

At a constant temperature and pressure, the first derivatives of ln K_{obs} by ln a_w or ln a_{Na+} are represented by Eqs. (21) and (22), respectively:

$$\frac{d\ln K_{obs}}{d\ln a_w} = -\left[\Delta n_w + \Delta n_{cs}\left(\frac{d\ln a_{cs}}{d\ln a_w}\right) + \Delta n_{Na^+}\left(\frac{d\ln a_{Na^+}}{d\ln a_w}\right)\right] \tag{21}$$

$$\frac{d\ln K_{obs}}{d\ln a_{Na^+}} = -\left[\Delta n_w\left(\frac{d\ln a_w}{d\ln a_{Na^+}}\right) + \Delta n_{cs}\left(\frac{d\ln a_{cs}}{d\ln a_{Na^+}}\right) + \Delta n_{Na^+}\right] \tag{22}$$

When the molarities of water and the cosolute are changed by altering the NaCl concentration, the activity coefficients are 1.0, and Eq. (22) can be rewritten as follows: [27,43]

$$\frac{d\ln K_{obs}}{d\ln[\mathrm{NaCl}]} = -\left[\Delta n_w\left(\frac{d\ln[H_2O]}{d\ln[\mathrm{NaCl}]}\right) + \Delta n_{cs}\left(\frac{d\ln[\mathrm{CS}]}{d\ln[\mathrm{NaCl}]}\right) + \Delta n_{Na^+}\right] \tag{23}$$

where $[H_2O]$, [CS], and [NaCl] are the molar concentrations of water, the cosolute, and NaCl, respectively.

4. Effect of cosolutes on binding of water to nucleic acids

The molecular crowding effects induced by cosolutes on DNAs with different structures (Fig. 3) were previously described. In a typical experiment, the stability changes of DNA duplex (D1) and quadruplex (Q1) in the absence and presence of various amounts (10, 20, 30, and 40 wt%) of PEG 200 (PEG with average molecular weight of 200) were measured. For a typical experiment, 5 μM D1 and 5 μM Q1 were dissolved in buffer [25]. The UV absorbance was measured with a Shimadzu 1700 spectrophotometer (Shimadzu, Kyoto, Japan) equipped with a temperature controller. Melting curves of D1 and Q1 were obtained by measuring the UV absorbance at 260 [10,44] and 295 [45] nm, respectively, in buffers containing 100 mM NaCl, 10 mM Na_2HPO_4 (pH 7.0), and 1 mM Na_2EDTA in the absence and presence of cosolutes. The heating rate was 0.5 °C min^{-1} for both D1 and Q1 because neither the shapes of the melting curve nor the melting temperatures (T_m) were affected by heating rates between 0.1 and 0.5 °C min^{-1}. Before the measurement, the sample was heated to 80 °C, gently cooled at a rate of 2 °C min^{-1}, and incubated at 0 °C for 30 min.

The normalized melting curves in solutions containing 0–40 wt% PEG 200 are shown in Fig. 4. These curves demonstrate that the absorbance of D1 and of Q1 changes at 260 and 295 nm as temperature increases in both the presence and absence of PEG 200. The T_m for 5 μM of D1 decreased from 67.0 to 55.5 °C as the PEG 200 concentration was increased from 0 to 40 wt% (Fig. 4a). On the other hand, the melting temperature for 5 μM of Q1 increased from 24.1 to 36.6 °C as the PEG 200 concentration was increased from 0 to 40 wt% (Fig. 4b). These results demonstrated that PEG 200 whereas it stabilized the G-quadruplex.

To quantify the PEG effect on the DNA stability, the observed equilibrium constants for the formation of D1 and Q1 structures and the values of the water activity for the buffer were also measured in 0 to 40 wt% PEG 200 at a constant salt (cation) concentration. The values of ln K_{obs} for D1 and Q1 at 25 °C at different concentrations of PEG 200 are plotted on the y-axis and the values of ln a_w for each solution are plotted on the x-axis in Fig. 5. The stability (ln K_{obs}) of D1 decreased linearly with the decrease in ln a_w. In contrast, the stability (ln K_{obs}) of Q1 increased linearly with the decrease in ln a_w. Although the slope of the plot includes two variable terms responsible for the cosolute and cation bindings, the linear plot in Fig. 5 (described by Eq. (21)) suggests that these variable terms are insignificant, and the slope approximately equals the constant term of $-\Delta n_w$. The slopes of ln K_{obs} vs. ln a_w for the D1 and the Q1 were estimated to be 99.1 ± 9.9 and −60.3 ± 1.8, respectively. If the slope of the linear plot (as shown in Fig. 5) is positive, water molecules are taken up upon formation of the DNA structure. If the slope is negative, water molecules are released during the folding reaction. Thus, these values for slopes of ln K_{obs} vs. ln a_w correspond to taken up of 3.5 ± 0.4 and to release of 4.0 ± 0.1 water molecules per nucleotide upon formation of D1 and Q1, respectively. Our results indicate that a decline in the water activity due to the addition of cosolutes should disfavor formation of structures that require uptake of water molecules and favor formation accompanied by the release of water. The stability changes in the crowding conditions are observed not only for DNA structures but also for RNA structures. Thus, the stability of nucleic acid structures can be controlled by the addition of cosolutes that alter hydration.

5. Effect of cosolutes on cation binding to nucleic acids

It is possible that cosolutes change the affinity of cations for nucleic acids by reducing the dielectric constant of the solution. Previously we found that the number of sodium ions bound to a DNA

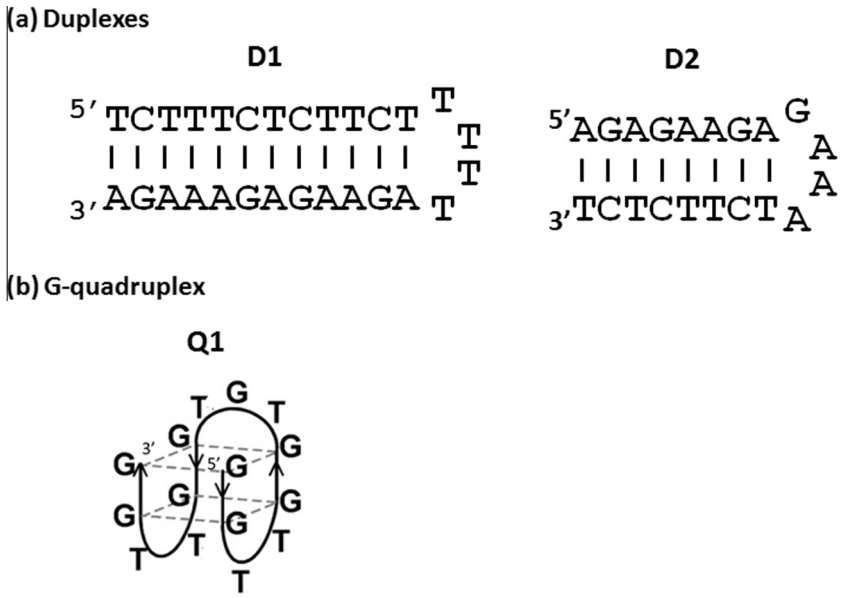

Fig. 3. Sequences and schematic structures of DNAs.

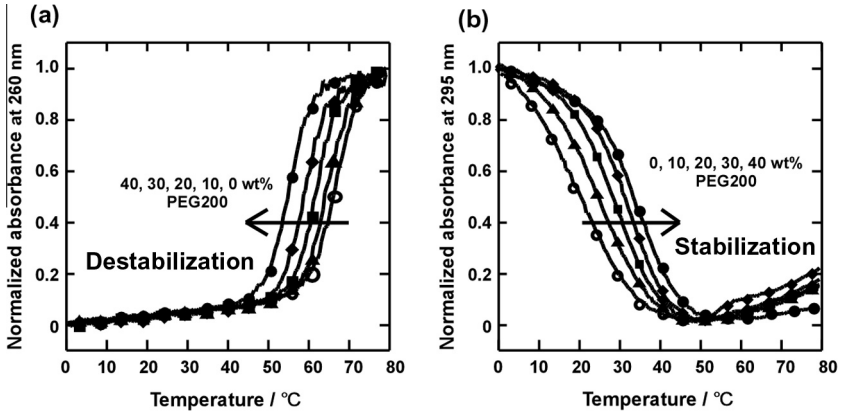

Fig. 4. UV melting curves of 5 μM (a) D1 and (b) Q1 in buffer containing 100 mM NaCl, 10 mM Na_2HPO_4 (pH 7.0), and 1 mM Na_2EDTA in the absence (open circles) and presence of 10 wt% (triangles), 20 wt% (squares), 30 wt% (diamonds), and 40 wt% (closed circles) PEG 200.

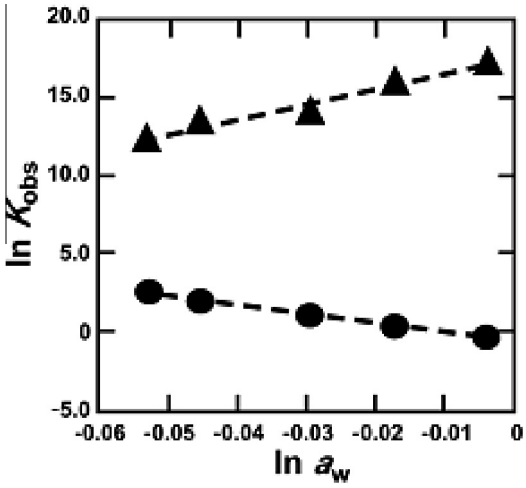

Fig. 5. Plots of ln K_{obs} vs. ln a_w for the formation of the D1 (triangles) and Q1 (circles) in a buffer of 100 mM NaCl, 10 mM Na_2HPO_4 (pH 7.0), and 1 mM Na_2EDTA containing 0, 10, 20, 30, or 40 wt% PEG 200 at 25 °C.

duplex decreases as the cosolute concentration increases. To investigate the effect of the cation concentration on the DNA stability under the molecular crowding conditions, we measured the UV melting curves for D2 at 10 or 600 mM NaCl in the presence and absence of 20 wt% PEG 200 or PEG 8000 (Fig. 2) [27]. In 600 mM NaCl the T_m decreased by 7.5 °C (from 70.8 to 63.3 °C) upon addition of PEG 200 and by 1.9 °C (from 70.8 to 68.9 °C) upon addition of PEG 8000. At 10 mM NaCl, the T_m decreased relative to that in 600 mM NaCl by 17.3 °C (from 70.8 to 53.5 °C) in the absence of PEG, by 13.6 °C (from 63.3 to 49.7 °C) in the presence of 20 wt% PEG 200, and by 11.3 °C (from 68.9 to 57.6 °C) in the presence of 20 wt% PEG 8000. Interestingly, PEG 8000 increases the thermal stability of the DNA duplexes at lower NaCl concentrations, whereas it decreases the stability at higher NaCl concentrations.

Using Eq. (23), we calculated the number of sodium ions bound $-\Delta n_{Na^+}$ upon D2 formation at 10, 20, 30, 40, and 50 wt% PEG 200 and 10 and 20 wt% PEG 8000. Fig. 6 shows plots of ln K_{obs} of D2 at 25 °C vs. ln [Na^+] at different concentrations of PEG. The stability of D2 (ln K_{obs}) increased linearly with the ln [Na+] between 10 and 600 mM NaCl. As described by Eq. (7), the slope of the plot includes a constant term, $-\Delta n_{Na^+}$, and two variable terms for water and cosolute binding, however, the linearity of the plots suggests that

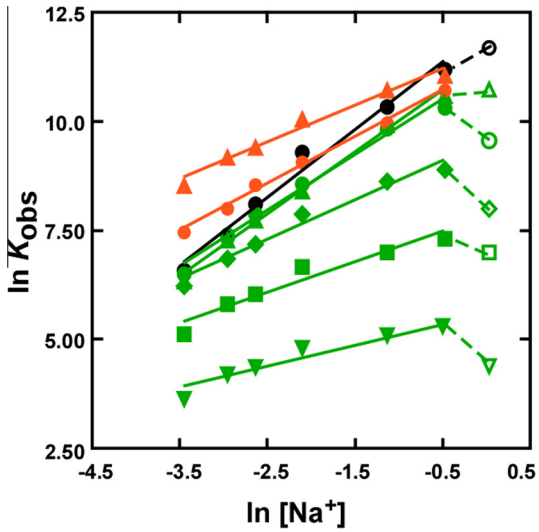

Fig. 6. Plots of ln K_{obs} for D2 at 25 °C vs. ln [Na^+]. The ln K_{obs} values at 25 °C were measured in solutions without PEG (black), with PEG 200 (green) at 10 wt% (triangles), 20 wt% (circles), 30 wt% (diamonds), 40 wt% (squares), or 50 wt% (inverted triangles) or with PEG 8000 (orange) at 10 wt% (triangles) or 20 wt% (circles). The data points that deviate from linearity are indicated with open symbols.

the slope is approximately equal to $-\Delta n_{Na^+}$ as applied for the determination of $-\Delta n_w$.

The slope of the plot in the absence of PEG was positive ($-\Delta n_{Na^+} > 0$), indicating that sodium ions bind during D2 formation. The $-\Delta n_{Na^+}$ value of 1.6 ± 0.1 indicates that 0.080 sodium ions bind per DNA phosphate, in agreement with previous reports that 0.020–0.075 sodium ions bind per phosphate to a DNA hairpin with a six base-pair stem and six loop nucleotides. The $-\Delta n_{Na^+}$ values for D2 in the presence of 10, 20, 30, 40, and 50 wt% PEG 200 were 1.4 ± 0.1, 1.2 ± 0.1, 0.91 ± 0.08, 0.70 ± 0.09, and 0.48 ± 0.09, respectively, and the values in the presence of 10 and 20 wt% PEG 8000 were 1.1 ± 0.1 and 0.84 ± 0.06, respectively, which were lower than the value in the absence of PEG. These results indicate that the number of sodium ions bound to D2 decrease as the PEG concentration is increased. Additionally, we found that PEG 8000 and PEG 200 had less effect on the number of water molecules bound upon duplex formation ($-\Delta n_w$) than on the number of cations bound [24]. These results suggest that PEG reduces the number of sodium ions but not the number of water molecules required for structure formation. Therefore, it is possible that DNA stability is controlled by the cosolutes and cation because cosolutes enhance the affinity of cation-nucleotide binding.

6. Size dependency of cosolute effects on nucleic acids

In order to analyze the cosolute effects on nucleic acids, we focused on the behavior of short strands of nucleic acid in the presence of small cosolutes. The lengths of the nucleic acid strands that are used in nanotechnology applications are generally under 50 nucleotides [1–9]. Cosolute effects on nucleic acids depend on the size of cosolutes and nucleic acids [24]. Studies of oligonucleotides that attempt to mimic the intracellular environment rely on the addition of neutral cosolutes to alter water activity. However, any cosolute affects both water activity and excluded volume. Typically, water activity is changed by the addition of small neutral organic compounds (e.g., alcohols, sugars, and ethylene glycol [EG]), whereas excluded volume is altered by addition of large polymers (e.g., PEG1000, PEG 10000, dextrans, and proteins). In both approaches, it is essential that the added cosolutes are inert and do not interact with either reactants or products of the reaction under study. Properly designed hydration or crowding studies typically use a number of different cosolutes to show that any observed effects are independent of the chemical properties of the individual cosolute additives. One early example studied hydration and crowding effects on the stability of DNA structures [16]. The small osmolytes EG, glycerol, acetamide, and PEG 200 were found to exert quantitatively similar effects: destabilizing duplex and stabilizing triplex and G-quadruplex with decreasing water activity [25]. In contrast, addition of the volume excluding cosolutes, including PEG 400, 1000, 8000 and 10,000, slightly stabilized duplexes and significantly stabilized triplex and G-quadruplex [46,47].

PEG may not be ideal for use in molecular crowding studies [48,49]. Crowding reagents should not bind to either the reactants or products of the reaction under study. Several studies have shown that the stabilization of protein structures by high molecular weight PEGs cannot be described quantitatively in terms of excluded volume alone; there is an attractive interaction between PEG and non-polar or hydrophobic side chains on the protein surfaces. Moreover, high molecular weight PEGs were recently shown to stack on G-quartets in DNA G-quadruplexes [48]. These attractive interactions compromise crowding experiments that employ high molecular weight PEG, although PEG 200 has been used as a cosolute (Fig. 4) and shown to stabilize G-quadruplex via decreasing the water activity. Quantitative data demonstrated that the effects of the polymers could be correlated with geometric models for excluded volume, and that their effects were due primarily to crowding [50]. In this case, effect of water activity and excluded volume changes on nucleic acids could be clearly separated. Thus, we should consider mainly a decrease of the water activity when the solution contains PEGs below about 200 molecular weight, but the excluded volume and direct interactions of higher molecular weight PEG likely affects structure formation of nucleic acids.

7. Preferential interaction between cosolutes and nucleic acids

Hydration is a key factor in the stability of structures formed by short strands of DNA so that small cosolutes including low molecular weight PEGs can be useful for regulating the stability of nucleic acids via changing the water activity without excluded volume effects. We estimated the number of water molecules bound during structure formation ($-\Delta n_w$) under the assumption that cosolutes are uniformly dispersed in a solution. However, one must consider the possibility of interactions between cosolutes and nucleic acids.

Although the water activity of solution is decreased by addition of small cosolutes, small cosolutes affect the structure formation of nucleic acid because of favorable or unfavorable chemical interactions of cosolute with the nucleic acid residues exposed on the surface or buried in the process of structure formation. There is a theory for preferential interactions between cosolutes and biopolymers [51,52]. Water molecules form hydration shells around biopolymers in solution. The structure of the hydration shell is changed by the local accumulation or exclusion of cosolutes at surface of biopolymers when cosolutes are added. For example, glycine betaine, urea, and guanidine hydrochloride can preferential bind to a biopolymer [53,54]. In contrast, hydrophilic cosolutes such as polyols, sugars, and amino acids are excluded from surface of biopolymers and induce the preferential hydration of biopolymers because the cosolutes increase the surface tension around the biopolymer in accordance with Gibbs' adsorption theory [55]. The degree of preferential interactions depends on accessible surface area (ASA) of the cosolutes and biopolymers which depends in turn on the hydrophobic and hydrophilic groups at the surface of biopolymers [52,56]. Taking account preferential interactions,

we can understand in detail the behavior of nucleic acids in the presence of cosolutes.

8. Cosolutes change the stability, structure, and function of nucleic acids

Formation of nucleic acid structures is accompanied by the formation of a hydrogen-bonded network of water surrounding the nucleic acid surface, [57] and the formation of the water network is highly sensitive to the water activity of the solution [16,38,41]. The stability of nucleic acid structures in the presence of small cosolutes is primarily affected by the degree of hydration of nucleic acid structures (as measured by $-\Delta n_w$), because the water activity of a solution is decreased by the presence of cosolutes. A decline in the water activity is unfavorable for formation of structures that require uptake of water molecules and favorable for structure formation accompanied by the release of water. For example, use of PEGs, polyols, and polysaccharides as cosolutes decrease the thermal stability of long and short DNA duplexes, because these cosolutes affect nucleotide hydration and reduce the water activity of a solution [16,19,25]. Also, alcohols, such as ethanol and propanol, decrease the stability of duplexes [58]. In contrast, the formation of triplexes, G-quadruplexes, i-motifs, and Holiday junctions involve the release of water and thus are stabilized in the presence of cosolutes [16,25,59–61]. The efficiency of stability changes by cosolutes strongly depends on cosolute structure. The efficiency is reduced when the cosolute has hydroxyl groups because hydroxyl groups interact with DNA reducing hydration (see section 7) [24,25].

PEG 8000 increases the thermal stability of the DNA duplexes at low NaCl concentrations, whereas it decreases the stability at high NaCl concentrations [27]. The difference in the duplex stabilities in the absence and presence of this cosolute increases when the NaCl concentration is lower than the physiological salt concentration. Moreover, the presence of PEG 8000 decreases the number of sodium ions bound $-\Delta n_{Na^+}$, but has less of an effect on the number of water molecules bound upon duplex formation ($-\Delta n_w$). It was reported that the addition of EG reduces the number of sodium ions bound to calf thymus DNA due to reduction of the dielectric constant of the solution [17]. It is assumed that the $-\Delta n_{Na^+}$ reflects not only the sodium ions tightly bound to nucleotide bases but also those bound through weak electrostatic interactions with phosphates, which exchange rapidly with those in the bulk solution. According to simple polyelectrolyte theory, $-\Delta n_{Na^+}$ is related to changes in the charge density or the axial spacing of nucleotide phosphate groups. When the charge density of nucleotide phosphates increases, cations bind more tightly. We found that the conformation of single-stranded DNA oligomers in the presence of PEG 200 or PEG 8000 differs from that in a solution without PEG, although the addition of PEG does not change the DNA duplex structure [62]. Therefore, it is possible that PEGs enhances the affinity of cations for nucleotides by reducing the dielectric constant. Moreover, the excluded volume effect on the nucleotides by high molecular weight PEGs also affect the cation binding. This observation suggests that cosolute in living cells offsets perturbations in salt concentration levels. Buffering the effects of changes due to environmental salt concentration favors the maintenance of nucleotide structures and cellular metabolism. Buffering of metabolism may also be mediated by the thermodynamics of biomolecular reactions. The concentration of proteins inducing molecular crowding inside living cells is known to vary with the cell cycle [63]. This might impact stability or the type of structure adopted by DNA. Accordingly, quantitative analyses of the biomolecular reactions in the presence of cosolutes can shed light on mechanisms of reactions occurring in living cells.

By employing the intracellular system using appropriate cosolutes, we can control the stability and structure of nucleic acids. For example, the G-quadruplex formed by the *Tetrahymena* telomere sequence undergoes a structure transition from anti-parallel form to parallel form upon addition of cosolutes because the parallel form, [26] which is more hydrated than anti-parallel form, is more stable than anti-parallel form in the presence of cosolutes [64]. Based on cosolute effects on nucleic acids, we can also enhance the functions of nucleic acids. When the hammerhead ribozyme changes conformation from the inactive to the more compact active form, which is capable of a cleavage reaction, water is likely to be released and metal ions are bound. We found that the activity of the hammerhead ribozyme is enhanced 20-fold in a solution containing 20 wt% PEG 8000 compared to that without PEG [65]. It was also reported that the activity of hairpin ribozyme is increased in the presence of neutral cosolutes, [66] supporting the hypothesis that the active form is less hydrated than the inactive conformation. The folded structure of group I ribozyme, which is more compact than the unfolded (inactive) form, is also stabilized in the presence of PEG; the dominant influence of the PEG in this case is likely the excluded volume effect [67]. Cosolutes can also be used to control the binding affinity of ligands to DNA. The binding of the thrombin aptamer to thrombin (a protein) involves uptake of water molecules, and thus the binding affinity is reduced by the addition of cosolutes [68,69]. In contrast, PEG increases the affinity of DNA polymerases for DNA because the binding is accompanied by dehydration [70]. In addition, several ligands that bind to the major groove of the DNA duplex have higher affinity in the presence of cosolute than in a dilute solution [71]. The presence of cosolutes mimics the beautiful system in cells and will be useful not only for understanding nucleic acid structures and functions in cells but also for the design of oligonucleotides to use the DNA nanotechnology.

9. Troubleshooting

The results of the experiments described herein are interpreted under the assumption that the DNA structures are the same in the absence and presence of cosolutes. CD spectroscopy and native gel electrophoresis are useful for the estimation of DNA structures. Before conducting hydration and cation binding analysis, the DNA structures in the presence and absence of cosolutes should be confirmed. Moreover, the slope of the ln K_{obs} versus ln [Na^+] plot has an inflection point (Fig. 5). Although the $-\Delta n_{Na^+}$ value in the region with a linear slope (−3.5 to −0.5 on the *x*-axis in Fig. 5), is constant, $-\Delta n_{Na^+}$ varies in the nonlinear region (more than −0.5 on the *x*-axis), likely due to increasing non-specific Na^+ binding to the bases at very high salt concentrations.

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